Unit 2 Quiz C: Coaching Guide

Don't just check the letter. For each question, answer the 'Ask yourself' prompts BEFORE you read the steps.

Q	1	2	3	4	5	6	7	8	9	10	11	12	13
Key	**B**	**C**	**B**	**B**	**A**	**C**	**C**	**A**	**A**	**B**	**A**	**C**	**B**

- **Score it:** right + sure = solid. Right + guess = review. Wrong = full walkthrough + Go deeper.
- Re-attempt every miss on paper tomorrow, without looking. If you get it right then, it sticks.

Question 1

2.1 Center of mass

A uniform 2.0 m rod of mass 4.0 kg has a 2.0 kg ball fixed to its right end. How far from the LEFT end is the system's center of mass?

ASK YOURSELF FIRST

- Where can you treat all of a uniform rod's mass as sitting?
- Should the answer be left or right of 1.0 m? Is it all the way out at 2.0 m?

STEP BY STEP

1. A uniform rod acts like a point mass at its center: 4.0 kg at x = 1.0 m (a line of symmetry).
2. The ball: 2.0 kg at x = 2.0 m.
3. x_{cm} = (4.0·1.0 + 2.0·2.0) / 6.0 = 8.0/6.0 ≈ 1.33 m.

ANSWER B. 1.33 m

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** Ignored the ball.
- **C:** Averaged 1.0 and 2.0 without weighting by mass.

GO DEEPER Where should the rod rest on a single finger to balance? (At the center of mass, 1.33 m from the left.)

Question 2

2.4 Equilibrium (2-D)

A sign of weight W hangs from two cables. Each cable makes 30° with the horizontal. What is the tension in EACH cable?

ASK YOURSELF FIRST

- The cables are nearly horizontal. Can a mostly sideways cable hold up much weight?
- Only the VERTICAL parts of the tensions hold the sign up. How big is each vertical part?

STEP BY STEP

1. Equilibrium. Horizontal: the two horizontal parts cancel (symmetry), so both tensions are equal.
2. Vertical: 2T sin 30° = W.
3. sin 30° = 0.5, so 2T(0.5) = W → T = W. Each cable carries the full weight of the sign!

ANSWER C. T = W

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** W/2 is only right if the cables are vertical (θ = 90°).
- **B:** Used cos 30° instead of sin 30°. Check which part is the vertical one.

GO DEEPER Sketch T vs. θ. As θ → 0, T → ∞. That's why you can never pull a clothesline perfectly straight with laundry on it.

Question 3

2.3 Newton's 3rd law

On ice skates, you push on a wall and glide away from it. Which force accelerates you?

ASK YOURSELF FIRST

- To accelerate YOU, the force has to act ON you. Which of these forces acts on you, and which acts on the wall?

STEP BY STEP

1. Your push acts on the wall, so it goes on the wall's FBD.
2. Its 3rd-law partner, the wall's normal force on you, acts on you and points away from the wall.
3. That's the only horizontal force on you (ice ≈ frictionless), so it causes your acceleration.

ANSWER B. The wall's force on you

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** That force acts on the wall.
- **C:** Internal forces can't accelerate a system's center of mass. You need an outside push.
- **D:** Inertia explains why you keep gliding AFTER you let go, not how you started.

GO DEEPER Would this work in deep space, pushing off a floating astronaut? (Yes. You'd both accelerate in opposite directions, with accelerations in the inverse ratio of your masses.)

Question 4

2.5 Connected systems

A 2.0 kg block and a 3.0 kg block on a frictionless floor are tied together with a light string. A 20 N force pulls the 3.0 kg block. What is the tension in the string?

ASK YOURSELF FIRST

- Which block does the string pull? Only that block needs to be isolated.
- The string accelerates the 2 kg block. Does it need the whole 20 N for that?

STEP BY STEP

1. System for a: $20 = (2.0 + 3.0)a \rightarrow a = 4.0\ m/s^2$.
2. Isolate the 2.0 kg block: the only horizontal force is T. $T = 2.0 \times 4.0 = 8.0$ N.
3. Check with the 3.0 kg block: $20 - 8.0 = 12 = 3.0 \times 4.0$ ✓.

ANSWER B. 8.0 N

WHY THE WRONG ANSWERS TEMPT YOU

- **C:** 12 N is the NET force on the 3 kg block, not the tension.
- **D:** The string doesn't carry the full pull. Part of it accelerates the front block.

GO DEEPER If you pulled the 2.0 kg block instead (from the other side), what would the tension be? (3.0 × 4.0 = 12 N. The string tension depends on what's behind it.)

Question 5

2.5 Force → motion graphs

A cart starts at rest on a frictionless track. The graph shows the net force on it. Which describes the cart's velocity?

ASK YOURSELF FIRST

- Constant force means constant what?
- When the force switches off, does the cart stop? (1st law!)

STEP BY STEP

1. 0–3 s: constant F → constant a = F/m → v grows linearly (a straight line up).
2. After 3 s: F = 0 → a = 0 → v stays at whatever it reached. It doesn't drop.

ANSWER A. Linear increase, then constant

WHY THE WRONG ANSWERS TEMPT YOU

- **B:** Treats the force graph like a velocity graph.
- **C:** The 'objects naturally stop' myth. Without friction, nothing slows the cart.

GO DEEPER Mass 2.0 kg: what speed does it reach? (a = 2.0 m/s², so v = 6.0 m/s.) Sketch x vs. t too: a parabola, then a straight line.

Question 6

2.6 Mass vs weight

A 70 kg astronaut stands on the Moon (g_{Moon} = 1.6 N/kg). What are her mass and weight there?

ASK YOURSELF FIRST

- Which one depends on where you are, mass or weight?

STEP BY STEP

1. Mass is the amount of matter, so it's the same everywhere: 70 kg.
2. Weight is the gravitational force: $W = mg_{Moon} = 70 \times 1.6 = 112$ N.

ANSWER C. 70 kg and 112 N

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** Used Earth's g.
- **B:** Divided the mass. Mass doesn't change.
- **D:** The Moon has gravity, just weaker.

GO DEEPER Inertia: would she find it easier to push a 100 kg crate sideways on the Moon than on Earth (frictionless floor both times)? (No. Same mass, same F for the same a.)

Question 7

2.6 Free fall

An astronaut on the Moon (no air) drops a hammer and a feather from the same height at the same time. They land together. The best explanation:

ASK YOURSELF FIRST

- Is the gravitational force on the hammer bigger or smaller than on the feather?
- Then why don't they get different accelerations?

STEP BY STEP

1. $F_g = mg$: the hammer is pulled MUCH harder.
2. But $a = F/m = mg/m = g$: the hammer also needs more force to accelerate, by exactly the same factor.
3. The mass cancels, so both fall at $g_{Moon} = 1.6$ m/s². (Apollo 15 did this in 1971.)

ANSWER C. Force ∝ mass, so the accelerations are equal

WHY THE WRONG ANSWERS TEMPT YOU

- **B:** The forces are very different. Only the accelerations match.
- **D:** Half right, but vague. Name the actual cancellation: a = mg/m.

GO DEEPER This 'mass cancels' idea also explains why $\mu_k = \tan\theta$ and why $v_{max} = \sqrt{(\mu g r)}$ on a curve. Spot it in other problems.

Question 8

2.7 Friction (stacked)

A small block rests on top of a larger block. A rope pulls the bottom block so that BOTH move together at CONSTANT velocity. What is the friction force on the top block?

ASK YOURSELF FIRST

- Draw the FBD of the TOP block only. Is it accelerating?
- If there were friction on it, which other horizontal force would balance that friction?

STEP BY STEP

1. Top block: constant velocity → $\Sigma F = 0$.
2. Horizontally, the only possible force is friction from the bottom block. Nothing else touches it sideways.
3. So friction must be 0. (If the rope ACCELERATED the blocks, static friction would point forward and drag the top block along.)

ANSWER A. Zero

WHY THE WRONG ANSWERS TEMPT YOU

- **B:** That's the MAXIMUM static friction, and also the 'motion needs a force' myth.
- **C:** Kinetic friction needs sliding between the surfaces, and these two move together.
- **D:** The rope doesn't touch the top block.

GO DEEPER Now the rope accelerates the pair at 2.0 m/s². What's the friction on a 1.0 kg top block, and which way? (2.0 N forward. Static friction is what accelerates it.)

Question 9

2.7 Friction + angles

You move a sled across rough snow with the same force F at the same angle θ, either PULLING (rope angled up) or PUSHING (handle angled down). Which gives the larger acceleration?

ASK YOURSELF FIRST

- Write F_N for each case, using the vertical part of F.
- Friction = μF_N. Which case has less friction?

STEP BY STEP

1. Pulling: $F_N = mg - F \sin\theta$. Pushing: $F_N = mg + F \sin\theta$.
2. Friction $\mu_k F_N$ is smaller when pulling.
3. Same forward part $F \cos\theta$ in both cases, less friction when pulling → bigger net force → bigger a.

ANSWER A. Pulling

WHY THE WRONG ANSWERS TEMPT YOU

- **B:** Grip doesn't help here. Friction opposes the sled's sliding.
- **C:** The horizontal parts are equal, but the friction isn't.

GO DEEPER Write a in symbols for the pulling case: $a = [F \cos\theta - \mu_k(mg - F \sin\theta)] / m$. Check: θ = 0 gives $(F - \mu mg)/m$ ✓.

Question 10

2.5 Modified Atwood

Block m_1 sits on a frictionless table. A string runs over an ideal pulley at the edge to block m_2, which hangs. What is the acceleration of the blocks?

ASK YOURSELF FIRST

- Treat both blocks as one system moving along the string. Which force drives it? Which masses have to be accelerated?
- Check your answer with limits: $m_1 \to 0$? $m_2 \to 0$?

STEP BY STEP

1. Driving force along the string: $m_2 g$. (m_1's weight is balanced by the table.)
2. Mass being accelerated: $m_1 + m_2$. The tension is internal, so it cancels.
3. $a = m_2 g / (m_1 + m_2)$.
4. Limits: $m_1 \to 0$ gives $a \to g$ (free fall ✓). $m_2 \to 0$ gives $a \to 0$ ✓.

ANSWER B. $m_2 g / (m_1 + m_2)$

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** Forgot that m_2 also has to be accelerated.
- **C:** That's the hanging Atwood. Here, m_1's weight doesn't pull along the string.
- **D:** Only true if $m_1 = 0$.

GO DEEPER Find T in symbols. ($T = m_1a = m_1m_2g/(m_1 + m_2)$, which is less than m_2g. It has to be, or m_2 wouldn't accelerate down.)

Question 11

2.9 Vertical loop

A 50 kg rider goes through the top of a loop of radius 10 m at 14 m/s, upside down. What is the normal force from the seat on the rider?

ASK YOURSELF FIRST

- Upside down at the top, which way is the center of the loop from the rider? Which way does the seat push?
- Is mv^2/r bigger or smaller than mg here?

STEP BY STEP

1. The center is DOWN from the rider. Both mg and F_N (seat pushing on the upside-down rider) point down.
2. Down = toward the center: $F_N + mg = mv^2/r$.
3. $mv^2/r = 50 \times 14^2/10 = 980$ N, and $mg = 490$ N.
4. $F_N = 980 - 490 = 490$ N, downward.

ANSWER A. 490 N, downward

WHY THE WRONG ANSWERS TEMPT YOU

- **B:** Added mg instead of subtracting it (that's the bottom of the loop).
- **C:** The seat is above the upside-down rider, so it can only push down.
- **D:** Zero only happens at exactly $v = \sqrt{(gr)} \approx 9.9$ m/s.

GO DEEPER What does the rider 'feel' compared with normal weight? ($F_N/mg = 1$: they feel exactly their normal weight, but upside down!)

Question 12

2.9 + 2.7 Rotor ride · BOSS

In a 'rotor' ride, riders stand against the wall of a spinning cylinder of radius 2.5 m, and the floor drops away. $\mu_s = 0.40$ between the rider and the wall. What is the minimum wall speed that keeps riders from sliding down?

ASK YOURSELF FIRST

- Which force points toward the center? Which force holds the rider up?
- Faster spinning gives a bigger normal force. Why does that help?

STEP BY STEP

1. Toward the center: the wall's normal force. $F_N = mv^2/r$.
2. Vertical: static friction up balances mg. To avoid slipping, $f \le \mu_s F_N$, so $mg \le \mu_s mv^2/r$.

3. $v_{min} = \sqrt{(gr/\mu_s)} = \sqrt{(9.8 \times 2.5 / 0.40)} = \sqrt{61.25} \approx 7.8$ m/s.

4. The mass cancels, so heavy and light riders stick equally well.

ANSWER C. 7.8 m/s

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** $\sqrt{(\mu gr)}$ is the flat-curve formula. Here friction is vertical and the normal force is the centripetal one.
- **B:** 4.9 = g/2. A guess.
- **D:** Forgot the square root (61 is v^2).

GO DEEPER Draw the FBD: three forces (mg down, f up, F_N toward the center). No 'centrifugal' arrow pushing outward. That feeling is your inertia.

Question 13

2.8 Springs · optional: ask if springs are on the test

A spring stretches 0.20 m when a 2.0 kg mass hangs from it on Earth. The same spring and mass are taken to the Moon (g = 1.6 N/kg). The stretch on the Moon is:

ASK YOURSELF FIRST

- The spring responds to the force pulling on it. Which force is that: mass, or weight?

STEP BY STEP

1. $kx = mg$, so $x \propto g$.
2. $x_{Moon} = 0.20 \times (1.6/9.8) \approx 0.033$ m.

ANSWER B. About 0.033 m

WHY THE WRONG ANSWERS TEMPT YOU

- **A:** A spring scale measures weight (force), not mass.
- **C:** Ratio upside down.

GO DEEPER Which would read the same on the Moon, a spring scale or a balance scale? (The balance, since it compares masses.)

Error log: Quiz C

For each miss or guess: what went wrong, using one of these types: **C** concept (didn't know the physics) · **R** representation (FBD/graph wrong) · **M** math/algebra · **F** FRQ wording (right idea, missing the words) · **T** time.

Q	Type	What I did vs. what I should have done	Re-try date / ✓

Score: _____ / 13 Sure-and-right: _____ Time used: _____